

Ki Score for East Africa:

An AI-Powered Decision-Support System for Inclusive Agricultural Credit

Kaleidofin | Digital Extension for Accountable Service Delivery
Enabling access to finance for everyone, everywhere

01 The Scale of the Problem

50 million Smallholder farmers in sub-Saharan Africa	52% of rural employment they represent in SSA	< 3% of bank lending in SSA reaches agriculture
--	---	--

This is not a demand problem. Smallholder farmers want credit and advisory services. Lenders and institutions want to serve them. What is missing is the decision-support layer that gives institutions the data, confidence, and tools to do so — quickly, fairly, and at scale.

The problem runs deeper than the absence of formal finance. **Even modern fintech approaches — mobile money scoring, credit bureau models, bank statement analysis — still fail to reach the vast majority of smallholder farmers.** These farmers have no bureau record, no M-Pesa or mobile money transaction history significant enough to score, and no bank account. By every existing measure, they are invisible to the financial system — not because they are poor credit risks, but because no data framework has been built to assess them using what actually exists about their farms, their land, and their environment.

97% of smallholder farmers in Africa have no formal financial safety net — not because they are undeserving, but because no system has ever been designed to assess and serve them using the data that already exists about their lives and farms.

02 Why Institutions Cannot Serve Smallholder Farmers Today

Agricultural lenders and extension institutions across East Africa face four structural barriers that prevent them from delivering accountable, inclusive, and effective services to the farmers they are meant to serve.

No Data at the Point of Decision Loan officers make judgement calls with almost no structured information on the farmer's actual risk profile. Climate exposure, crop history, soil conditions, seasonal cashflow — none of this feeds into the decision. Even fintech tools that use bureau or mobile money data fail here: most smallholder farmers have no bureau record and insufficient M-Pesa or bank transaction history to generate a meaningful score.	Decisions Are Slow and Inconsistent Manual underwriting takes days to weeks, is inconsistent across officers, and creates no audit trail. Many creditworthy farmers are turned away not because they are risky, but because the institution cannot assess them. The result is systematic exclusion of the very segment these institutions exist to serve.
Geographic and Climate Risk Is Invisible A farmer in a drought-prone lowland zone and one in a stable highland are treated identically. Without geographic risk differentiation, institutions over-lend in high-risk zones and under-lend in safe ones — harming both farmers and portfolio quality.	No Early Warning — Only Post-Default Reaction Institutions have no way to see portfolio stress building until it has already materialised as defaults. By then it is too late to intervene — seasonal windows have closed, farmers have already suffered, and recovery is slow and expensive.

03 Introducing Ki Score: A Decision-Support System Built for East Africa

Ki Score is an AI-powered decision-support system that brings climate, soil, crop, and farm-level data to the point of lending decisions — enabling institutions to serve smallholder farmers with speed, accuracy, and accountability.

Critically, Ki Score can generate a robust risk assessment using **publicly available alternate data alone** — no bureau record, no M-Pesa history, no bank statement required. Bureau and transaction data can be layered in to **enhance** the score further, but they are not prerequisites. This is what makes Ki Score structurally different from every other credit tool in this market.

PRIMARY INPUTS (always available)	INTELLIGENCE ENGINE	SERVICE DELIVERY OUTPUT
Core alternate data — sufficient for decisioning: - Climate & weather signals (CHIRPS, ERA5, SPEI) - Soil & land characteristics	- Multi-level data fusion — individual + geographic signals - Seasonal feature engineering — long rains / short rains windows	- Risk tier for each farmer or loan application - Climate-driven risk rationale — transparent & auditable

(SoilGrids — 250m resolution) • Rural socioeconomic context (CIAT/GRDI) • Farm profile: crop type, acreage, irrigation practice • Geographic & agro-ecological zone data • Individual farmer demographic data Enhancement inputs (when available): • Credit bureau loan history • Mobile money / M-Pesa transactions • Bank statement data • Savings transaction history	• Anomaly scoring vs 10-year climate baseline • AI/ML risk segmentation — interpretable, explainable • Point-level extraction at farm GPS coordinates • Temporally anchored — no data leakage at origination • 600,000+ East Africa loan performance records in training • 100,000+ individual KALRO farmer records (Kenya)	• Recommended lending terms (LTV, tenor, covenants) • Portfolio-level early warning signals • Embedded via API or no-code business rules • No technical overhead for the local institution
--	--	---

For the first time, a lender can make a data-driven credit decision on a farmer they have never lent to — with no bureau file, no mobile money record, no bank statement. Ki Score makes thin-file and no-file farmers visible to the financial system. This is the new era of inclusive agricultural finance.

04 A New Era of Inclusive Agricultural Finance

Traditional and fintech credit tools share the same fundamental limitation: they require a financial footprint. Bureau history. Mobile money transactions. Bank statements. For smallholder farmers in East Africa, none of these exist at sufficient scale to support a credit decision.

TRADITIONAL LENDING	FINTECH / MOBILE MONEY SCORING	KI SCORE
Requires: • Bank account and statement history • Formal credit bureau record • Collateral or guarantor	Requires: • Significant mobile money transaction history • Smartphone and digital literacy	Requires: • Farm GPS location • Basic farmer demographic and farm profile data • Crop type, acreage, irrigation practice

Result: 97% of smallholder farmers excluded - No decision possible for first-time borrowers	- Consistent mobile network access Result: - Still excludes farmers with thin mobile money footprints - Urban-biased; rural smallholders underserved	Result: - Any farmer, anywhere can be assessed - Bureau / M-Pesa data enhances score further — but not required - Thin-file and no-file farmers become creditworthy
---	--	--

The Paradigm Shift

From: "We cannot lend because we have no financial data on this farmer."

To: "We can lend, because we know the farmer's climate exposure, soil profile, crop type, and local risk context."

05 The Data Stack: What Makes Ki Score Genuinely Different

Most credit and advisory systems in this market rely on a single satellite signal or a basic demographic form. Ki Score fuses five climate datasets with real loan performance at scale — enabling a level of individual risk differentiation that was previously impossible for this borrower segment.

Source	What It Captures	Resolution	Role in Model
CHIRPS	Rainfall, precipitation anomaly	5 km, monthly	Drought and flood stress signal — seasonal and anomaly-based
ERA5-Land	Temperature, surface runoff, vegetation index	9 km, hourly	Climate zone classification; waterlogging and heat stress
SoilGrids	Sand %, clay %, organic carbon, pH, coarse fragments	250 m	Agro-ecological zone proxy; drainage and water retention capacity
CIAT / GRDI	Global Rural Deprivation Index	10 km	Socioeconomic vulnerability overlay — credit risk context
SPEI	Drought severity index, 1–48 month scales	55 km	Slow-onset stress signal; lagged drought impact on repayment

100,000+ deidentified individual farmer records from KALRO Kenya — the only large-scale individual-level agricultural dataset of its kind in East Africa.

600,000+ actual loan performance records across 5 East Africa securitisation transactions — real credit behaviour, not survey proxies or synthetic data.

Credit bureau data combined with climate signals is currently being integrated as the next model layer — early results show meaningful additional risk separation beyond climate data alone.

06 How Ki Score Segments Risk: From Data to Actionable Decisions

Ki Score does not produce a single opaque score — it produces an interpretable, explainable risk tier grounded in multiple data dimensions. The segmentation framework draws on climate, geographic, soil, farm-level, and individual farmer data to differentiate risk in ways that no single-signal model can achieve.

The factors the model draws on include: climate variability (rainfall, temperature, drought indices); geographic data on farm profiles including crop type, acreage, and yield patterns; soil types and their interaction with climate and hazard variables; irrigation practice and land-use characteristics; and individual farmer demographic data. Using this combination, Ki Score can generate a risk decision even for a farmer with no bureau history and no mobile money footprint. Bureau and transaction data, when available, are layered in as additional signal — they enhance the model, they are not the foundation of it.

The following illustrates the kind of segmentation our Kenya agricultural portfolio modelling produces:

Ki Score differentiates risk not just by geography, but by crop type and loan structure. In our Kenya portfolio, maize farms under severe rainfall stress show a **50% higher default rate** than those with adequate rainfall, while cabbage farms in the same conditions show only **6% increased risk**. Similarly, high temperature combined with longer tenors amplifies risk significantly — maize farms with 36-month loans in high-temperature zones show **40% higher default rates** than equivalent shorter-tenor loans. This means Ki Score can inform not just the lending decision, but the specific product structure — tenor, LTV, pricing — appropriate for each farmer's climate context.

LOW RISK (~20% of portfolio)	MEDIUM RISK (~50% of portfolio)	HIGH RISK (~30% of portfolio)
Typical Profile - Highland climate zone, stable rainfall - Low drought stress index (SPEI near-normal)	Typical Profile - Moderate climate variability - Elevated seasonal runoff or rainfall anomaly	Typical Profile - Warm lowland zone, high rainfall variability - Rocky or acidic soils — poor water retention - High drought severity or flood risk signal

• Good soil drainage and water retention • Low surface runoff anomaly What the Model Enables • Standard underwriting with preferential terms • Higher loan-to-value, longer tenors • More creditworthy farmers approved — inclusion gains	• Sparse prior-season vegetation signal • Borderline SPEI readings What the Model Enables • Standard terms with monitoring covenants • Seasonal check-ins at critical growth stages • Early alert triggers if conditions deteriorate	• Lagged vegetation stress from prior season What the Model Enables • Tighter LTV, shorter tenors • Mandatory early warning monitoring • Prevention — not reaction — to default
--	---	---

Ki Score is an inclusion tool, not an exclusion tool. For every risk tier — including the most climate-exposed geographies — it recommends a credit product structure that makes lending viable, not one that closes the door. A farmer in a high-risk warm lowland zone should not be turned away; they should receive a loan with appropriate tenor and monitoring. The goal is to bring more farmers into the financial system with terms that reflect their actual situation.

07 What Accountable Service Delivery Looks Like in Practice

Track 3 calls for decision-support systems that improve timeliness, transparency, and inclusion in service delivery. A critical part of inclusion is ensuring that data-driven tools do not inadvertently replicate bias — that farmers in climate-stressed geographies receive appropriate credit, not blanket rejection. Ki Score is designed with this in mind: every output is a recommendation for how to serve a farmer, not whether to serve them. Here is what that looks like concretely:

Dimension	Without Ki Score	With Ki Score
Decision speed	Days to weeks — manual review at every level	Real-time automated response at point of application
Basis for decision	Loan officer intuition, inconsistent across officers	Climate + soil + behavioural data — auditable and consistent
Inclusion	Farmers with no bureau record, no M-Pesa history, and no bank account are systematically excluded — even by modern fintech tools. These thin-file and	Thin-file and no-file farmers can now be decisioned using alternate data alone. Bureau and mobile money data enhance the score further — but are not

	no-file farmers have no financial data to score against.	required. Inclusion expands by 20–30%.
Explainability	No reason given to farmer for rejection or pricing	Clear risk tier with climate and behavioural rationale
Geographic fairness	One-size-fits-all pricing regardless of actual local risk	Differentiated by actual climate zone and soil profile
Portfolio oversight	Reactive — institutions only know about stress after default	Predictive early warning signals — intervene before defaults materialise

08 Proven at Scale: The Track Record Behind Ki Score

\$7B+ credit underwritten using Ki Score globally	10M+ customers across 60 partnerships in multiple countries	20–30% boost in farmer approval rates	< 3% PAR 90 — defaults stay low as volumes grow	5+ years market-tested including COVID waves
---	---	---	--	--

In India - Models trained on data from 40 million+ unique customers - Deployed operationally for agri loan underwriting at scale - Proven through multiple credit cycles and macroeconomic stress periods 3–5x lift in customer approvals vs baseline decisioning	In East Africa - PD model live across 5 securitisation transactions — 600,000+ loan records - Validated against actual repayment outcomes, not synthetic proxies - Readiness level 7.5 — validated in real-world uncontrolled conditions - Member of the World Bank One Million Farmer Platform — building agri credit decisioning for Kenya's largest agri programme
---	---

Kaleidofin operates across India, Bangladesh, Kenya, Uganda, and Nigeria. The East Africa adaptation is underway, not hypothetical — built on real loan data, real climate signals, and real institutional relationships already in place.

What the Data Shows: Climate Signals Separate Credit Risk

These are not modelled assumptions — they are observed default rate differentials from our Kenya agricultural lending portfolio:

Indicator	Low / Moderate Stress	Severe Stress	Risk Increase
Temperature hazard (ERA5)	0.31 default rate	0.37 default rate	+19%
Waterlogging hazard (surface runoff)	0.28 default rate	0.36 default rate	+29%
Rainfall hazard (CHIRPS)	0.26 default rate	0.36 default rate	+38%
Poverty index (CIAT/GRDI)	0.29 default rate	0.37 default rate	+28%

Climate data alone produces statistically meaningful risk separation. Combined with credit bureau data and individual farmer records, the separation strengthens further — and the model continues to improve as more East Africa data accumulates.

09 The East Africa Opportunity: Why Now, Why Here

Why Kenya first

- Highest agri-fintech intensity in Africa — 30% of the continent's agri-tech startups operate here
- Existing infrastructure: SACCOs, MFIs, bureau data, mobile money — the distribution already exists
- World Bank Million Farmer Program covers 1 million farmers across 12 crop and 11 livestock value chains
- KALRO dataset provides 100,000+ individual farmer records not available in any other East African market
- Kaleidofin has live operations in Nairobi and active institutional relationships

Why the region is replicable

- Rwanda, Tanzania, Uganda share similar climate, crop calendar, and institutional profiles to Kenya

Target geographies

Country	Phase
Kenya	Primary — Phase 1
Rwanda	Expansion — Phase 2
Tanzania	Expansion — Phase 2
Uganda	Expansion — Phase 2

Addressable scale

- 50M+ smallholder farmers in sub-Saharan Africa
- \$430B estimated shortfall in financial services to smallholders globally

- | | |
|---|---|
| <ul style="list-style-type: none"> • CHIRPS, ERA5, and SoilGrids cover all four markets at the same resolution — model retraining, not rebuilding • SACCO and cooperative infrastructure across the region provides a ready distribution channel • Existing East Africa securitisation data spans multiple countries — model has cross-border signal | <ul style="list-style-type: none"> • < 3% of SSA bank lending currently reaches agriculture |
|---|---|

10 Kaleidofin: Built for This Problem

Who We Are

Kaleidofin is a multi-product fintech dedicated to financial inclusion, founded in 2017 in Chennai, India. We began East Africa operations in Nairobi in 2023.

Mission: Enabling access to finance for everyone, everywhere.

- Offices in Chennai, Bangalore, and Nairobi
- Operations across India, Bangladesh, Kenya, Uganda, and Nigeria
- 60+ institutional partnerships across 5 countries
- 98% of customers are women; 60% are marginal farmers
- Member of the World Bank One Million Farmer Platform

Our Products

Ki Score

Supervised AI/ML credit decisioning tool — trained on 40M+ customer records. Tailored to partner portfolios. Embeds into lending flows via low-code APIs or no-code business rules. Boosted approval rates by 20–30% with PAR 90 < 3%.

Ki View / Ki Monitor

Portfolio monitoring and early warning system. Predictive models that surface stress signals before defaults materialise. Roll rate analysis, delinquency profiling, covenant management, and waterfall analysis for lenders.

\$7B+

credit underwritten
using Ki Score

10M+

customers served
globally

60+

institutional
partnerships

98%

customers are
women

60%

are marginal
farmers

11 Pilot Plan: A Phased Approach to East Africa Deployment

PHASE 1 — KENYA PILOT	PHASE 2 — REGIONAL EXPANSION	PHASE 3 — SCALE & INSTITUTIONALISE
<i>Months 1–6</i> • Calibrate Ki Score for East Africa agri credit context • Deploy across a pilot cohort of lenders and advisory institutions in Kenya • Integrate with existing SACCO and MFI workflows via API • Measure: how it improves local institution via approval rate lift, default rate by risk tier, institution feedback on decision quality • Establish baseline for inclusion and accountability metrics	<i>Months 6–18</i> • Extend to Rwanda, Tanzania, and Uganda • Adapt seasonal windows and climate features to local crop calendars • Leverage existing Kaleidofin institutional relationships for distribution • Deepen model with in-country loan performance data and credit bureau integration — combining climate signals with bureau data is already showing promising results in early testing, pointing to significantly sharper risk differentiation in the next model version • Introduce cashflow risk dimension as savings transaction data becomes available	<i>Months 18+</i> • Embed Ki Score as a standard decision-support layer across agri lending in East Africa • Explore integration with government extension systems and DFI programmes • Open the risk framework to additional data sources — weather index triggers, satellite yield estimates • Establish Ki Score as the regional benchmark for agri credit decisioning

The infrastructure to reach millions of smallholder farmers already exists — through SACCOs, MFIs, cooperatives, and agri-lenders across East Africa. Ki Score is the intelligence layer that makes it work.

12 Team

TBD	TBD	ABHIJIT	ADHIP
Tech	Product (or) Business	Head of Data Science	Data Scientist
TBF	TBF	TBF	TBF

Kaleidofin — founded 2017, Chennai, India. Operations in India, Bangladesh, Kenya, Uganda, and Nigeria. Member of the World Bank One Million Farmer Platform. \$7B+ credit underwritten using Ki Score globally.

12 The Ask: What a Partnership with M4D Enables

We bring a validated, deployed decision-support system and deep East Africa roots. M4D brings the network, credibility, and programme infrastructure to take this to the scale it deserves.

What Kaleidofin Brings

- Validated decision-support system — readiness level 7.5
- 600,000+ real East Africa loan performance records
- 100,000+ individual KALRO farmer records (Kenya)
- \$7B+ credit underwritten globally using Ki Score
- Live East Africa operations in Kenya, Uganda, and Nigeria
- World Bank One Million Farmer Platform membership
- Proven model architecture deployable across Kenya, Rwanda, Tanzania, Uganda

What M4D Partnership Enables

- Phase 1 (\$10K): Pilot deployment and calibration in Kenya — establishing baseline metrics on inclusion, decision speed, and accuracy
- Phase 2 (\$100K): Full East Africa rollout with pilot and implementation support across Rwanda, Tanzania, Uganda
- Phase 3 (\$250K): Scale to institutional adoption — embedding Ki Score as the standard agri credit decision layer across the region
- Network access: IFAD and CGIAR connections to bring in MFI, government extension, and development finance partners who can amplify reach
- Co-investment: M4D visibility and credibility to attract private and public sector co-investment for sustained scale
- Endorsement: M4D backing accelerates institutional adoption in a market where development organisation credentials matter

"For the first time, every smallholder farmer — regardless of their financial footprint — can be seen, assessed, and served. Ki Score makes that possible."